

Architecture Decision Record: ADR-005

Configuring Avahi for the Local Zenoh Router Domain

Abel-Raul Mazilu

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Status: Accepted
Project: DVerse Communication Platform 2.0

1 Context

The DVerse Communication Platform 2.0 uses a local Zenoh router to support communication between nodes in the deployment environment. Nodes must be able to discover and connect to this router reliably without depending on a changing IP address.

Using raw IP addresses creates operational and security problems. Local network addresses may change across restarts, router reconfiguration, DHCP lease renewal, or deployment to another network. If certificates are issued against an IP address that later changes, nodes may fail validation or require unnecessary certificate replacement.

To avoid this, the local Zenoh router should be reachable through a stable local domain name. Avahi can provide this through multicast DNS (mDNS), allowing the router to advertise a predictable `.local` hostname on the local network. This gives other nodes a stable name to connect to and gives the team a semi-permanent domain name target for certificate issuance.

2 Decision

Sprint 4 will configure Avahi to advertise the local Zenoh router's address through mDNS. The Zenoh router will be assigned a stable local hostname, and nodes will connect to the router using that hostname rather than a changing IP address.

The implementation will include the following capabilities:

- Configure Avahi on the machine hosting the local Zenoh router.
- Define a stable `.local` hostname for the Zenoh router.
- Broadcast the Zenoh router address through mDNS so local nodes can resolve it automatically.

- Update node configuration to use the advertised hostname instead of a hard-coded IP address.
- Ensure certificate generation and validation can target the stable hostname.

The expected result is that nodes can resolve and connect to the Zenoh router through a semi-permanent local domain name, improving reliability and certificate stability across local network changes.

3 Alternative Decisions

One alternative was to continue using hard-coded IP addresses for the Zenoh router. This was rejected because IP addresses can change and are not a reliable long-term identity for local routing or certificate validation.

Another alternative was to require manual host-file entries on every node. This was rejected because it creates operational overhead, is error-prone, and does not scale well as nodes are added or replaced.

A third alternative was to rely on external DNS. This was rejected for the local network use case because the router should be discoverable inside the local deployment environment without requiring external DNS infrastructure.

4 Consequences

4.1 Positive

- Nodes can discover the local Zenoh router through a stable local domain name.
- Certificate issuance can target a hostname rather than a changing IP address.
- Local network changes are less likely to break node connectivity.
- New nodes can connect with less manual configuration.

4.2 Negative

- Avahi becomes part of the local deployment requirements.
- mDNS availability must be verified on the target network.
- Hostname conflicts must be avoided and documented.
- Troubleshooting must include both Zenoh connectivity and mDNS resolution checks.

Deliverable	Description
Avahi configuration	Configure Avahi on the host running the local Zenoh router.
Stable hostname	Define a predictable <code>.local</code> hostname for router discovery.
mDNS broadcast	Advertise the Zenoh router address to local nodes through mDNS.
Node configuration	Update nodes to connect using the advertised hostname instead of a raw IP address.
Certificate target	Use the stable hostname as the target identity for semi-permanent certificate issuance.

5 Scope

Statement on AI Use

Artificial intelligence was used to support the drafting of parts of this document. The generated text was subsequently reviewed, checked, and edited where necessary by me, the author, Abel-Raul Mazilu.